| | | |
|---|---|---|
| 1(a)(i) | 24 protons 28 neutrons 24 electrons | **1** |
| 1(a)(ii) | ($^{54}$Cr has two) more neutrons (than $^{52}$Cr) | **1** |
| 1(a)(iv) | The fourth isotope has any value of RIM less than 51.996 | **1** |
| 1(b)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ (3d^5\ 4s^1)$ | **1** |
| 1(b)(ii) | 6 | **1** |

# Examiner report missing
# 9701_w25_qp_23 Question 1

| | | |
|---|---|---|
| 2(a)(i) | **M1** nuclear charge increases (across period) **AND** similar shielding<br><br>**M2** greater / stronger attraction to nucleus | **2** |

| | | |
|---|---|---|
| 2(a)(ii) | **M1** A$l$ loses (3) electrons / A$l$ is 3+, and P gains (3) electrons / P is 3–<br><br>**M2** A$l^{(3+)}$ has (one) fewer shell (than $P^{(3-)}$) | **2** |

# Examiner report missing
# 9701_w25_qp_23 Question 2

| | | |
|---|---|---|
| 1(a)(i) | 25 protons 30 neutrons 25 electrons | **1** |
| 1(a)(ii) | (atoms of an element) with the same number of protons and different number of neutrons | **1** |
| 1(a)(iii) | **M1** (relative) isotopic mass.<br>**OR** (relative isotopic) mass of Mn-55 and Mn-53<br><br>**M2** (relative) abundance of each isotope | **2** |
| 1(b)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ (3d^5\ 4s^2)$ | **1** |
| 1(b)(ii) | 5 | **1** |

# Examiner report missing
# 9701_w25_qp_22 Question 1

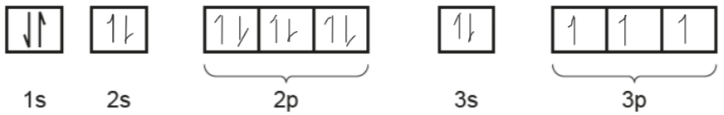
| | | |
|---|---|---|
| 1(a)(i) | 24 protons 28 neutrons 24 electrons | **1** |
| 1(a)(ii) | ($^{54}$Cr has two) more neutrons (than $^{52}$Cr) | **1** |
| 1(a)(iv) | The fourth isotope has any value of RIM less than 51.996 | **1** |
| 1(b)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ (3d^5\ 4s^1)$ | **1** |
| 1(b)(ii) | 6 | **1** |

# Examiner report missing
# 9701_w25_qp_21 Question 1

| | | |
|---|---|---|
| 2(a)(i) | **M1** nuclear charge increases (across period) **AND** similar shielding<br><br>**M2** greater / stronger attraction to nucleus | **2** |

| | | |
|---|---|---|
| 2(a)(ii) | **M1** A$l$ loses (3) electrons / A$l$ is 3+, and P gains (3) electrons / P is 3–<br><br>**M2** A$l^{(3+)}$ has (one) fewer shell (than $P^{(3-)}$) | **2** |

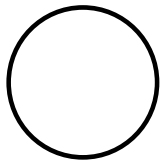
# Examiner report missing
# 9701_w25_qp_21 Question 2

| | | |
|---|---|---|
| 1(a) | <table><tr><th>*Mg*</th><th>*Si*</th><th>*P*</th></tr><tr><td>**M1** Giant metallic</td><td>**M2** Giant molecular</td><td>**M3** Simple molecular</td></tr></table> | **3** |
| 1(c)(i) | $Ar^{2+}(g) \rightarrow Ar^{3+}(g) + e^{(-)}$<br><br>**M1** a correct equation which represents an ionisation process *e.g.* $X^{n+} \rightarrow X^{(n+1)+} + e^{(-)}$<br>**M2** correct equation to show third ionisation energy of Ar including (g) state symbols | **2** |
| 1(c)(ii) | **M1** nuclear charge<br>**M2** distance of outer electron(s) from nucleus **OR** ionic radius **OR** shielding (by inner shells and / or sub-shells) | **2** |
| 1(c)(iii) | **M1** (most significant factor) = distance of outer electron(s) from nucleus **OR** ionic radius **OR** shielding **OR** shells<br><br>**M2** (3rd IE) value for (Na and) Mg is (much) greater than A*l* (to Ar)<br>**OR** (Na and) Mg has electron(s) removed from a shell that is closer to the nucleus / less shielding<br>**OR** (Na and) Mg has electron(s) removed from a shell that is lower in energy | **2** |
| 1(c)(iv) |  | **1** |

# Examiner report missing
# 9701_s25_qp_24 Question 1

| | | |
|---|---|---|
| 1(a)(i) | $1s^2\ 2s^2\ 2p^5$ | **1** |
| 1(a)(ii) | 2 (pairs) | **1** |
| 1(a)(iii) |  | **1** |
| 1(b)(i) | $S(g) \rightarrow S^+(g) + e^{(-)}$ **OR** $S(g) - e^{(-)} \rightarrow S^+(g)$ | **1** |
| 1(b)(ii) | **M1** pair of electrons in a 3p orbital (in S) repel each other<br>**M2** outweighs the effect of increase in nuclear charge | **2** |
| 1(b)(iii) | **M1** $Na^+ < Ne < F^-$<br>**M2** (all have) same number of electrons / isoelectronic<br>**M3** larger species has smaller nuclear charge / fewer protons<br>**M4** (so) smaller nuclear attraction (to outer electrons) | **4** |

# Examiner report missing
# 9701_s25_qp_23 Question 1

| | | |
|---|---|---|
| 2(a)(i) | **route 1**<br>**M1** average (weighted) mass of (an) atom(s) (of an element)<br>**OR** average (weighted) mass of the isotope(s) (of an element)<br><br>**M2** compared to the mass of the unified atomic mass unit<br>**OR** on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units)<br>**OR** on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units)<br>**OR** compared to $\frac{1}{12}$ mass of one atom of C-12<br><br>**route 2**<br>**M1** average mass of one mol of isotopes / atoms (of an element)<br><br>**M2** relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes<br>**OR** when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g | **2** |
| 2(b) | 1 (pair) | **1** |
| 2(c) |  | **1** |
| 2(d) | species present in an ion of 56-$Fe^{3+}$ : number<br>electrons : **M1** 23<br>**M2** neutrons : 30 | **2** |
| 2(e) | **M1** smaller / decreases<br><br>**M2** loss of outer / valence shell (of electrons)<br>**OR** loss of electrons so increase in (force of) attraction from nucleus<br>**OR** loss of electrons so increase nuclear attraction | **2** |

# Examiner report missing
# 9701_s25_qp_22 Question 2

| | | |
|---|---|---|
| 2(a) | atoms (of an element) with same number of protons and different number of neutrons | **1** |
| 2(b)(ii) | <table><tr><th>particle</th><th>number of particles in one atom of $^{56}Fe$</th></tr><tr><td>protons</td><td>26</td></tr><tr><td>nucleons</td><td>56</td></tr></table> | **2** |
| 2(c) | 5 | **1** |
| 2(d) | $Fe(g) \rightarrow Fe^{+}(g) + e^{(-)}$ | **1** |
| 2(e) | **M1** same electronic configuration **OR** same shielding<br>**M2** same number of protons **OR** same nuclear charge<br>**M3** so same nuclear attraction to outer shell electron<br>**M4** no change / will not be affected | **4** |

# Examiner report missing
# 9701_s25_qp_21 Question 2

| | | |
|---|---|---|
| 2(a) | electrostatic attraction<br><br>between positive metal ions / cations and delocalised electrons | **2** |

| | | |
|---|---|---|
| 2(b)(i) | ionic radius increases / ions larger (down the group)<br><br>extra shell (of electrons) | **2** |

| | | |
|---|---|---|
| 2(b)(ii) | ionisation energies increase because less shielding / successive ionic radii are smaller / greater attraction (of electron) to nucleus<br><br>(large difference between 2nd and 3rd IE)<br>because 3rd $e^-$ coming from an inner shell / lower energy level | **2** |

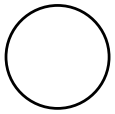
| | | |
|---|---|---|
| 2(c)(i) | most of the mass is in the nucleus<br><br>because protons and neutrons / nucleons are found in the nucleus | **2** |

| | | |
|---|---|---|
| 2(c)(ii) | | **2** |

| isotope | number of protons | number of neutrons |
|---|---|---|
| magnesium-24 | 12 | 12 |
| magnesium-25 | 12 | 13 |
| magnesium-26 | 12 | 14 |

| | | | |
|---|---|---|---|
| 2(c)(iii) | neutron beam | horizontal straight line from same source | **3** |
| | electron beam | downwards curving line from same source | |
| | | line deflected more than proton beam | |

# Examiner report missing
# 9701_m25_qp_22 Question 2

| | | |
|---|---|---|
| 1(a)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ | **1** |
| 1(a)(ii) |  | **1** |
| 1(a)(iii) | **3** (three) | **1** |
| 1(b) | | **3** |

| isotope | number of protons | number of neutrons | total number of electron shells |
|---|---|---|---|
| $^{103}_{45}Rh$ | • 45 | 58 | • 5 |
| $^{191}_{77}Ir$ | • 77 | • 114 | 6 |
| $^{193}_{77}Ir$ | • 77 | • 116 | 6 |

**Question 1**

**(a) (i)** Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

**(ii)** This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

**(iii)** Several candidates gave an answer of one, whilst many gave the correct answer.

**(b)** Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

**(c) (i)** The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

**(ii)** Many candidates incorrectly used all three isotopes to calculate the $A_r$ of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

**(d) (i)** Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of $H_2O$ to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

**(ii)** Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

**(iii)** Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra $CH_2$ units to the chain.

**iv)** The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

| | | |
|---|---|---|
| 2(a)(i) | chlorine = yellow-green<br>bromine = orange / brown / red<br>iodine = silver grey / black | **1** |
| 2(a)(iii) | green colour disappears | **1** |
| 2(a)(iv) | H—Hal covalent bond strength **decreases** down the group | **1** |

**Question 2**

**(a) (i)** The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

**(ii)** Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

**(iii)** Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

**(iv)** Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

**(b) (i)** Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

**(ii)** This question was well answered.

**(iii)** Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

**(c) (i)** Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

**(ii)** The definition of disproportionation was well answered by many candidates.

**(iii)** This equation was well known.

**(iv)** This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of $NO_2$ is also formed. Those candidates who described how the $Cl^-$ ion was formed, prior to its reaction with $Ag^+$ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form $Cl^-$. Many stated that the $Cl^-$ was contained in $CHCl_3$, and not a product from its hydrolysis.

| | | |
|---|---|---|
| 3(a)(i) | <table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table> | **3** |

| | | |
|---|---|---|
| 3(c)(i) | **M1** general increase | **1** |
| | **M2** increase in IE between IE**4** and IE**5** is noticeably the largest increase between IE's shown | **1** |

| | | |
|---|---|---|
| 3(c)(ii) | $Si^{+}(g) \rightarrow Si^{2+}(g) + e^{-}$ | **1** |

**Question 3**

**(a) (i)** This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

**(ii)** Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

**(iii)** Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

**(b) (i)** The most common errors were to give the formula for magnesium chloride as MgC$l$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

**(ii)** Increases was correctly stated by many candidates.

**(iii)** Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, $NH_4^+$, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

**(c) (i)** Correct completion of the graph was shown by many candidates.

**(ii)** Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

**(d) (i)** There were many correct candidate answers stating an increase in the number of electrons, from $CH_4$ to $PbH_4$, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

**(ii)** The shape of a molecule of $SiH_4$ was well answered.

**(e) (i)** Many correct answers for the equation were present. A common error was to use HC$l$ as a reactant, instead of C$l_2$.

**(ii)** Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiC$l_4$ is added to water.

**(iii)** Credit was given to candidates for recognising that the structure and bonding of solid, $SiO_2$, was giant covalent and liquid SiC$l_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in $SiO_2$ but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiC$l_4$ molecules. Several answers compared the strength of the covalent bonds in both $SiO_2$ and SiC$l_4$ and many answers incorrectly stated that the structure of $SiO_2$ contained double bonds between the Si and O atoms.

**(f)** Many candidates gave the correct answer.

| | | |
|---|---|---|
| 1(a)(i) | lowest energy state owtte<br><br>examples<br>• (with) low**est** energy<br>• (with) no external energy<br>• (with) no added energy<br>• (where) electrons are not promoted to a higher energy level<br>• most energetically stable | **1** |

| | | |
|---|---|---|
| 1(a)(ii) | [Ar] ↑ ↑ ↑ ☐ ☐ ↑↓<br>*The 3 × d electrons may go in any of the d orbitals* | **1** |

| | | |
|---|---|---|
| 1(a)(iii) | 12 | **1** |

1(b)(i)

| isotope | RIM | # protons | # neutrons |
|---|---|---|---|
| $^{93}_{41}Nb$ | 92.91 | • 41 | • 52 |
| $^{181}_{73}Ta$ | 180.95 | • 73 | • 108 |

**2**

**Question 1**

**(a) (i)** Understanding of 'ground state' was rarely described correctly. Confusion of 'ground state' with 'natural' or 'standard' state was seen.

**(ii)** Many candidates gave the correct representation of the arrangement of electrons in the 3d and 4s orbitals.

**(iii)** The most common incorrect answer was '6'; perhaps deduced by those who considered only one of the energy levels n = 2 or n = 3.

**(b) (i)** Most candidates gave the correct deduction of the number of protons and neutrons in each species.

**(ii)** Confusion of the terms 'relative atomic mass' with 'relative isotopic mass' was seen in weaker candidate responses. Comparisons of mass were often incomplete or incorrect, examples included no identification of the specific carbon isotope.

**(iii)** Fully correct answers used the relative isotopic mass given in Table 1.1 correctly and gave their final answer to 2 decimal places. Weaker responses did not select the relevant data from Table 1.1. Final answers were not always given to the 2 decimal places specified.

| | | |
|---|---|---|
| 3(a)(i) | (natural = )lightning | **1** |
| | (man-made = )internal combustion engines | **1** |

| | | |
|---|---|---|
| 3(c)(i) | $P^{2+}(g) \rightarrow P^{3+}(g) + e^{(-)}$ | **1** |

| | | |
|---|---|---|
| 3(c)(ii) | general increase | **1** |
| | increase in IE between IE5 and IE6 is noticeably the largest increase between IEs shown | **1** |

**Question 3**

**(a) (i)** Many candidate answers referred to lightning as responsible for one natural occurrence of $NO_2$. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

**(ii)** Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of $NO_2$ as a catalyst in the production of sulfur-based acid rain.

**(iii)** Candidates who were familiar with this area of the syllabus described how PAN forms from $NO_2$. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with $NO_2$.

**(iv)** Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of $H_2$ rather than $H_2O$.

**(v)** Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

**(b) (i)** Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

**(ii)** The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

**(iii)** Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

**(iv)** Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

**(c) (i)** Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

**(ii)** A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

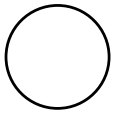
**(d)** Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

**(e)** Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

**(f) (i)** This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in $P_2$ identified as $sp^2$ or $sp^3$
- in $P_2$ overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

**(ii)** Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

| | | |
|---|---|---|
| 1(a)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ | **1** |
| 1(a)(ii) |  | **1** |
| 1(a)(iii) | **3** (three) | **1** |

1(b) — **3**

| isotope | number of protons | number of neutrons | total number of electron shells |
|---|---|---|---|
| $^{103}_{45}Rh$ | • 45 | 58 | • 5 |
| $^{191}_{77}Ir$ | • 77 | • 114 | 6 |
| $^{193}_{77}Ir$ | • 77 | • 116 | 6 |

**Question 1**

**(a) (i)** Most candidates were able to give a correct electronic configuration of: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. A few candidates included the $3d^74s^2$ given in the question. This was not required in the answer.

**(ii)** This question was well answered with most drawings attempting to represent a sphere as a circle. A few diagrams showed the x, y and z axes drawn to represent a 3D shape.

**(iii)** Several candidates gave an answer of one, whilst many gave the correct answer.

**(b)** Most candidate answers were correct for the total number of protons and neutrons, but the total number of electron shells, for rhodium, were often incorrectly given as 4.

**(c) (i)** The definition was not well answered. Many candidates stated that it was the average mass of an isotope/weighted average mass of an isotope or the average mass of isotopes. Very few answers gave a complete definition.

**(ii)** Many candidates incorrectly used all three isotopes to calculate the $A_r$ of iridium and divided by 100 or used the two isotopes of iridium but divided by 100 instead of by 50.

**(d) (i)** Many candidates were aware that to calculate an integer value of x, the water of crystallisation in $RhCl_3 \cdot xH_2O$, a mole ratio calculation was required. The ratio of $H_2O$ to $RhCl_3$ was given/deduced as 20.5 per cent : 79.5 per cent by mass, from which a mole ratio of 0.38 : 1.14 can be calculated. Many candidates chose a slightly different route and used the expression $18x \div (209.4 + 18x) = 0.205$, to correctly calculate a vale of $x = 3$.

**(ii)** Many candidate answers mentioned that stereoisomers contained the same 'molecular formula' but omitted the fact that they also contained the same 'structural formula', which is the key feature for this definition. Several candidates correctly stated that the stereoisomers had 'different arrangements in space' but made no mention of what was arranged differently in space e.g. atoms/groups.

**(iii)** Many candidates failed to recognise that a single product would be formed and gave explanations about adding extra $CH_2$ units to the chain.

**iv)** The diagrams of the two stereoisomers of but-2-ene were often correct. Errors included confusion on the position of the C=C, in the chain, on one or both structures and incorrect groups or incorrect arrangements.

| | | |
|---|---|---|
| 2(a)(i) | chlorine = yellow-green<br><br>bromine = orange / brown / red<br><br>iodine = silver grey / black | **1** |

| | | |
|---|---|---|
| 2(b)(i) | a species with one or more unpaired electrons | **1** |

**Question 2**

**(a) (i)** The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

**(ii)** Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

**(iii)** Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

**(iv)** Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

**(b) (i)** Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

**(ii)** This question was well answered.

**(iii)** Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

**(c) (i)** Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

**(ii)** The definition of disproportionation was well answered by many candidates.

**(iii)** This equation was well known.

**(iv)** This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of $NO_2$ is also formed. Those candidates who described how the $Cl^-$ ion was formed, prior to its reaction with $Ag^+$ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form $Cl^-$. Many stated that the $Cl^-$ was contained in $CHCl_3$, and not a product from its hydrolysis.

| | | |
|---|---|---|
| 3(a)(i) | <table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table> | **3** |

| | | |
|---|---|---|
| 3(c)(i) | **M1** general increase | **1** |
| | **M2** increase in IE between IE**4** and IE**5** is noticeably the largest increase between IE's shown | **1** |

| | | |
|---|---|---|
| 3(c)(ii) | $Si^{+}(g) \rightarrow Si^{2+}(g) + e^{-}$ | **1** |

**Question 3**

**(a) (i)** This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

**(ii)** Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

**(iii)** Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

**(b) (i)** The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

**(ii)** Increases was correctly stated by many candidates.

**(iii)** Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, $NH_4^+$, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

**(c) (i)** Correct completion of the graph was shown by many candidates.

**(ii)** Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

**(d) (i)** There were many correct candidate answers stating an increase in the number of electrons, from $CH_4$ to $PbH_4$, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

**(ii)** The shape of a molecule of $SiH_4$ was well answered.

**(e) (i)** Many correct answers for the equation were present. A common error was to use $HCl$ as a reactant, instead of $Cl_2$.

**(ii)** Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

**(iii)** Credit was given to candidates for recognising that the structure and bonding of solid, $SiO_2$, was giant covalent and liquid $SiCl_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in $SiO_2$ but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid $SiCl_4$ molecules. Several answers compared the strength of the covalent bonds in both $SiO_2$ and $SiCl_4$ and many answers incorrectly stated that the structure of $SiO_2$ contained double bonds between the Si and O atoms.

**(f)** Many candidates gave the correct answer.

| | | |
|---|---|---|
| 1(a) | <table><tr><td>$Mg^{2+}$</td><td>24</td><td>12</td><td>10</td></tr><tr><td>$Al^{3+}$</td><td>27</td><td>13</td><td>10</td></tr></table><br>4 correct = 2 marks<br>2 or 3 correct = 1 mark<br>0 or 1 correct = 0 marks | **2** |
| 1(b) | **M1** it / $Al^{3+}$ is smaller (compared to $Mg^{2+}$)<br>**M2** greater nuclear attraction for remaining electrons<br>**M3** same shielding effect AND greater nuclear charge | **3** |

**Question 1**

**(a)** The majority of answers were correct.

**(b)** Some excellent explanations were seen. Occasionally the aluminium ion was incorrectly described as larger than the magnesium ion. In general, the explanations focussed, at least in part, on the difference in nuclear attraction which arises as a result of the difference in nuclear charge and the same shielding effect of both species.

**(c)** Clearly labelled diagrams, which showed a regular arrangement of sodium cations surrounded by delocalised electrons, were seen in a significant proportion of answers. Poorer diagrams showed randomly placed positive ions or made no reference to 'delocalised' electrons. Occasionally, the positive symbols were described incorrectly as 'nuclei' or 'protons' or 'atoms'.

**(d) (i)** The majority identified the structure and bonding present in silicon. A much smaller proportion used this knowledge to explain why silicon had a high melting point in terms of the high energy requirement required to break many strong covalent bonds. A common misconception described intermolecular forces in a giant molecular structure.

**(ii)** Of those that annotated Fig.1.1, many correctly compared the melting points of P, S and C*l* with those of the three metals provided. Comparison of the melting points between P, S and C*l* was less well known.

**(e) (i)** This question proved to be demanding. The state of both sodium oxide and sulfur dioxide under room conditions was not well known. Misunderstanding of the term 'state of oxide' and oxidation number was seen on occasions. A common misconception described aqueous solutions of sodium oxide with a pH of 7.

**(ii)** Correct identification of this reaction as acid–base or neutralisation was common.

**(iii)** The product for the reaction of $P_4O_{10}$ with an excess of water was well known. Equations were not always balanced.

**(f) (i)** The majority of answers demonstrated a good understanding of the term amphoteric.

**(ii)** There was some confusion with this reaction and the reaction of aluminium oxide and sodium hydroxide. The correct formula of the product was not well known.

| | | |
|---|---|---|
| 3(a) | **M1** rates of forward and reverse / backward reactions are equal<br>**M2** no change in measurable properties **OR** concentration of reactants **AND** products remain constant | **2** |

| | | |
|---|---|---|
| 3(c) | *full electronic configuration of* $Fe^{3+}$ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$ | **1** |

**Question 3**

**(a)** Accurate descriptions of this definition were seen. Some weak responses were vague and incomplete, e.g., 'when forward and backward reaction is equal'. The misconception that dynamic equilibrium occurs only when 'concentration of reactants and products are the same' was also seen.

**(b) (i)** This question proved to be challenging. Many candidates were unable to apply le Chatelier's principle to this novel equilibrium reaction.

**(ii)** Some excellent answers, presented clearly and logically, were seen in this demanding question. A significant proportion of candidates confused the terms 'amounts' and 'concentrations'. Understanding of the different mathematical relationship between 2x and $x^2$ was not always appreciated. Correct deduction of the unit was common.

**(c)** Many answers were not based on the knowledge that 4s electrons are lost before 3d electrons when the electronic configuration of a transition metal cation is described.

**(d)** Correct dot-and-cross diagrams were seen. Weaker responses showed correct bonding electron arrangement and ignored the additional electron present in a $SCN^-$ anion.

| | | |
|---|---|---|
| 1(a)(i) | total # 3p $e^-$ 3 4 5 | **1** |
| | total # unpaired $e^-$ 3 2 1 | **1** |
| 1(a)(ii) | $P(g) \rightarrow P^+(g) + e^{(-)}$ | **1** |
| 1(a)(iii) | 1000 ($kJmol^{-1}$) | **1** |
| | (S) less / least / lowest nuclear attraction (to outer electrons) (than P **OR** C*l*)<br>**OR**<br>nuclear attraction for C*l* is stronger (compared to S (and or P)) | **1** |
| | **<u>S less nuclear charge than C*l*</u>**<br>OR<br>**<u>C*l* has a greater nuclear charge than S</u>** | **1** |
| | (**S** has a greater nuclear charge (of outer electrons) than **P BUT)**<br>S has two electrons in a (3)**p orbital AND** resulting in spin-pair repulsion | **1** |
| 1(b)(i) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ | **1** |
| 1(b)(ii) | • decreases (from $P^{3-}$ to C$l^-$)<br>• increased attraction (of outer $e^-$) for nucleus **owtte**<br>• increased nuclear charge<br>• same shielding<br>•√•√ | **2** |

**Question 1**

**(a) (i)** Most candidates were able to complete the table to show the number of electrons in the 3p subshells of P, S and Cl, as 3, 4 and 5 respectively.

The total number of unpaired electrons in these elements proved challenging for some candidates, Common incorrect answers were 1, 0 and 1 respectively.

**(ii)** The equation to represent the first ionisation of P was well answered. A common error was to omit the state symbols of (g) throughout.

**(ii)** The question required candidates to choose one of three possible values, given in the question, to represent the first ionisation energy of S and to explain their choice.

Most candidates did not mention that S has a pair of electrons in a 3p **orbital**, which experience spin-pair repulsion. This results in S having an ionisation energy value of 1000 kJ $mol^{-1}$, which is lower than both P and Cl. The explanation required candidates to recall that whilst S has a nuclear charge higher than P and lower than Cl, its nuclear attraction to the outer electrons is the lowest of these three elements.

Many candidates chose a value between P and Cl and supported their choice with a incorrect statement that the 'nuclear attraction to the outer electrons' increased across the period.

**(b) (i)** The full electronic configuration of the $P^{3-}$ ion was well answered as: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$.

**(ii)** The decrease in ionic radius from $P^{3-}$ to $S^{2-}$ and $Cl^-$ was well answered. Many correct explanations included mention of an increased attraction of the outer electron to the nucleus resulting from an increased nuclear charge with each ionic species having the same shielding.

**(c)** Observations were required for three tests carried out on separate samples of NaCl(aq).

When $Br_2$(aq) is added to a sample no reaction occurs, and the colourless solution of NaCl(aq) turns orange (because of the added $Br_2$).

Most answers stated that the observation from the addition of concentrated $H_2SO_4$ to NaCl(aq) is misty/steamy fumes, not recognising that the NaCl is in aqueous solution (and not a solid). The solution remains colourless as there is no visible reaction.

A white precipitate is formed when $AgNO_3$(aq) is added to NaCl(aq).

**(d) (i)** This question provided information that the chemical behaviour of $POCl_3$ is similar to that of $PCl_5$; $POCl_3$ has a low melting and boiling points and reacts vigorously with water suggesting that the structure and bonding of $POCl_3$ is simple covalent.

A small number of candidates gave a good explanation for this information and appreciated that the '*simple*' structure of $POCl_3$ was related to weak intermolecular forces of attractions and that the '*covalent*' nature of the compound is linked to the *hydrolysis* reaction with water.

**(ii)** The equation for the reaction of $POCl_3$ with water was generally quite well answered. Errors included an incorrect formula for $H_3PO_4$ and/or incorrect ratio between the water molecules and HCl product.

**(iii)** Most candidates were awarded full marks for the dot-and-cross diagram. A common error was the use of a dative bond between the P and O atoms.

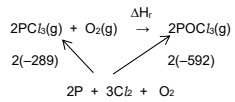
**(e) (i)** Most candidates were awarded the first mark for the definition of the enthalpy change of formation, for stating 'enthalpy change when 1 mole of a compound is formed'. However, very few correctly stated the second requirement that it is 'from its elements in their standard states'.

Many answers mentioned standard conditions which was not a requirement.

**(ii)** The calculation of the bond energy of P=O (in $PCl_3$), using $\Delta H_f$ and bond energy ($O_2$) data, proved challenging for most candidates.

Use of incorrect energy cycles was the major problem and full marks were rarely awarded.

A correct energy cycle is:




**$\Delta H_r$ = 2(–592) – 2(–289) = -606 kJmol$^{-1}$**

| bonds broken | bonds formed |
|---|---|
| ~~6 P-Cl~~ | ~~6 P-Cl~~ |
| 1 O=O (496) | 2 P=O |

496 - 2P=O = -606
-2P=O = -1102
**P=O = 551kJmol$^{-1}$**

| | | |
|---|---|---|
| 1(a)(i) | decreases (from Si to S) | **1** |

| | | |
|---|---|---|
| 1(b)(i) | total $e^-$ in s subshell 6 6 6 | **1** |
| | total $e^-$ in p subshell 8 9 10 | **1** |

| | | |
|---|---|---|
| 1(b)(ii) | $Si(g) \rightarrow Si^+(g) + e^-$ | **1** |

| | | |
|---|---|---|
| 1(b)(iii) | 1060 (is circled) | **1** |
| | P / it has greater attraction of nucleus for outer electrons compared to Si<br>ORA | **1** |
| | P / it has greater nuclear charge than Si<br>ORA | **1** |
| | S has two electrons in a (3)p orbital<br>**AND**<br>resulting in spin-pair repulsion | **1** |

**Question 1**

**(a) (i)** The variation in atomic radius from silicon to sulfur was, in general, well known.

**(ii)** Many candidates appreciated that the differences in melting points were a consequence of the differences in structures. Identification of the types and relative strengths of the forces of attraction broken during melting were referred to less often. Common misconceptions included the presence of intermolecular forces in silicon or did not refer to intermolecular forces, neither in general terms, nor as instantaneous dipole induced dipole attractions in sulfur.

**(b) (i)** Most candidates correctly calculated the number of electrons in p subshells.

**(ii)** Many equations showed appropriate species. State symbols, essential in any equation to represent the first ionisation energy process, were included less often.

**(iii)** Candidates with an excellent knowledge of the trend in ionisation energy across a period identified the correct ionisation energy for phosphorus. They explained both the general trend and the effect of spin pair repulsion on a pair of electrons in a p orbital of sulfur compared to phosphorus. Weaker candidates either identified the value which represents a general trend in ionisation energy across the period and explained this trend or identified the correct value for ionisation energy with details that only explained why sulfur had a lower first ionisation energy than phosphorus.

**(iv)** This was answered well by most candidates.

**(v)** Appropriate description of the shape of an $SCl_2$ molecule as ‘bent’ or ‘v-shape’ was seen often. ‘Linear’ was a common incorrect answer.

| | | |
|---|---|---|
| 2(a) | curved line from (0,0) to reach same horizontal line | **1** |

| | | |
|---|---|---|
| 2(e)(i) | species with 1 or more unpaired electrons | **1** |

**Question 2**

**(a)** Some candidates appeared to confuse graphs of rate versus time with those showing concentration versus time.

**(b)** Well-rehearsed arguments related the significance of increasing temperature on the position of equilibrium to the example given. More able candidates showed no ambiguity and made explicit reference to the effect of an increase in temperature on the position of this equilibrium in terms of either the '*forward reaction is exothermic'* or '*the reverse reaction is endothermic'*. Less precise answers included '*the reaction moves backwards as it's an exothermic reaction'*.

**(c)** This question proved to be demanding. A few candidates found $K_p$ using an appropriate method; they worked out the mole fraction of the components, converted them into partial pressures and then substituted these values into the correct expression for $K_p$. Weaker candidates calculated $K_p$ directly from moles and included moles in the units. Deduction of units also presented difficulties. Many answers gave units in terms of Pa with no attempt to convert kPa to Pa within the calculation.

**(d)** Lightning was well known as a natural process responsible for formation of atmospheric NO and $NO_2$. 'Catalytic converter' as the man-made process was a common incorrect answer.

**(e) (i)** Many correct definitions of the term free radical were seen. Ambiguous descriptions referred to '*unpaired electron'* as '*unshared*' or '*unbonded*' or '*free*'. Common incorrect answers referred to a free radical as *'a species with a lone pair of electrons'*.

**(ii)** The pair of equations which showed the catalytic role of $NO_2$ in the oxidation of $SO_2$ were not well known. The equation to represent the regeneration of $NO_2$ appeared to cause most difficulties with many incorrectly balanced equations seen.

**(iii)** The environmental consequence of the oxidation of sulfur dioxide was well known.

**(f) (i)** Identification of a magnesium containing base was relatively common. Incorrect answers included salts, rather than bases, which contained magnesium or bases that were not made from magnesium. The formula of the compound identified was not always correct.

**(ii)** This equation proved challenging for many. Some answers incorrectly balanced equations which included all correct species. Use of an incorrect formula for magnesium nitrate was a common error, including $Mg_3N_2$.

**(iii)** Identification of the change in thermal stability down Group 2 nitrates was fairly well known.

| | | |
|---|---|---|
| 1(a)(i) | total # 3p $e^-$ 3 4 5 | **1** |
| | total # unpaired $e^-$ 3 2 1 | **1** |
| 1(a)(ii) | $P(g) \rightarrow P^+(g) + e^{(-)}$ | **1** |
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**Question 1**

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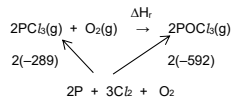
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| bonds broken | bonds formed |
|---|---|
| ~~6 P-Cl~~ | ~~6 P-Cl~~ |
| 1 O=O (496) | 2 P=O |

496 - 2P=O = -606
-2P=O = -1102
**P=O = 551kJmol$^{-1}$**

| | | |
|---|---|---|
| 1(a)(i) | giant metallic | **1** |

| | | |
|---|---|---|
| 1(d)(iii) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9(4s^0)$ | **1** |

**Question 1**

**(a) (i)** A high proportion of candidates knew that copper is metallic. Many candidates omitted that the lattice is giant.

**(ii)** Diagrams showing the metallic bonding as an array of positive metal ions surrounded by delocalised electrons were varied, with many either omitting or not correctly labelling the positive ions and delocalised electrons.

**(b) (i)** Some candidates incorrectly referred to a mole of $^{12}C$ atoms or just an atom of $^{12}C$.

**(ii)** Many correct answers were seen. Some candidates omitted the word 'average'.

**(iii)** The question was quite well answered. The major error included the use of 63 and 65 as the masses of the copper isotopes instead of the data given in the question.

**(c)** Very few candidates recognised that the ion with an abundance of 23% and an *m/e* of 31.5 was that of $^{63}Cu^{2+}$.

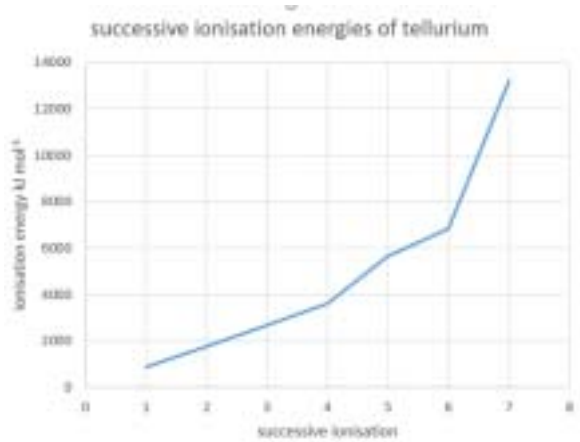
**(d) (i)** The equation was not well answered overall. Few candidates recognised that it was a redox reaction. Many answers contained electrons on one side of the equation and/or were completely unbalanced.

**(ii)** The identification of $Cu^{2+}$ as the oxidising reagent in this reaction was generally not well known. Many candidates answered with just 'copper'.

**(iii)** The full electronic configuration of $Cu^{2+}$ was frequently incorrect with the $4s^2$ electrons still present in many answers.

| | | |
|---|---|---|
| 1(a) | | **3** |

| | nucleon number | number of neutrons | number of electrons |
|---|---|---|---|
| Tellurium-130 | 130 | 78 | 52 |

| | | |
|---|---|---|
| 1(b) | 1 s | **1** |
| 1(c) | $Te(g) \rightarrow Te^{+}(g) + e^{(-)}$ | **1** |
| 1(d)(i) | *any two factors responsible for increase in IE in first 6 IE from:*<br>• (increased) **nuclear** attraction / attraction for **nucleus**<br>• (decreased) shielding (by sub-shells) (IE 4 to 5)<br>• spin-pair repulsion (IE 1 to 2) | **2** |
| 1(d)(ii) |  <br>**M1** general increase<br>**M2** jumps from 4 to 5 and (largest) 6 to 7 | **2** |

**Question 1**

**(a)** This was generally well answered. Some candidates were not familiar with the term nucleon number, often interpreting this as neutron number.

**(b)** Many candidates answered this question correctly. Common errors included not specifying **1**s and giving 5p as the answer.

**(c)** Candidates generally answered this well. State symbols were not always included, especially in $Te^{+}$.

**(d) (i)** The majority of candidates correctly identified two factors. Many candidates incorrectly stated that the shielding effect remains constant. An 'increase in nuclear charge' was a common incorrect response that negated other descriptions.

**(ii)** Most candidates recognised a general increase in successive ionisation energies, with a majority of those able to indicate the most significant jump from the $6^{th}$ to $7^{th}$ ionisation energy. Many answers incorrectly included a graph with a dip, most often in energy at the $6/7^{th}$ ionisation energy.

**(e)** Candidates who correctly referenced spin-pair interactions gained credit. Few candidates could use the VSEPR theory correctly.

**(f) (i)** This was generally not answered well. Many responses included a mixture of numbers and '*x*'s. A small proportion of candidates gave a correctly balanced equation.

**(ii)** Many candidates correctly identified this as a 'hydrolysis' reaction. A common answer was 'displacement', which was not enough to gain credit.

| | | |
|---|---|---|
| 1(a)(i) | power of an atom to attract electrons to itself | **1** |
| 1(c)(i) | $H(g) \rightarrow H^+(g) + e^-$ | **1** |
| 1(c)(ii) | H (cannot undergo second ionisation because it only) has one electron / $H^+$ has no electron | **1** |
| 1(c)(iii) | $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ | **1** |

**Question 1**

This question tested ideas of atomic structure, periodicity and some quantitative chemistry. Knowledge of electronegativity was less secure and many Period 3 equations were incorrectly presented as redox reactions.

**(a) (i)** There were some very good answers here, although many candidates tried to answer in terms of ionisation or electron affinity.

**(ii)** Few candidates were awarded both marks as this required a balance of points to construct an argument. Spin-pair repulsion arguments were unnecessarily presented on a number of occasions. Many mentioned the distance of the outermost electron from the nucleus without any reference to its significance. Other answers included non-justified references to ionic radius or ionisation energy.

**(b) (i)** A whole array of different diagrams was seen, with only the ionic form ($Li^+$ and $H^-$) acceptable. All electrons, only four in total, were required to be shown.

**(ii)** Linear, trigonal planar and planar were popular incorrect answers.

**(c) (i)** The most common error here was the omission of state symbols from the elemental species. Many answers suggested that $H_2$ was being ionised.

**(ii)** This was well answered by many candidates.

**(iii)** Most candidates answered this well, although $3p^6$ was commonly seen.

**(d) (i)** A large number of answers erroneously suggested $H_2SO_4$ as a product.

**(ii)** Both $Na_2S$ and $Na_2SO_4$ appeared regularly, as did $H_2$. Candidates are advised to consider whether redox has occurred in these reactions.

**(iii)** Many incorrect versions for the formula of the carbonate were seen such as $Mg(CO_3)_2$ or $MgCO_2$. Many candidates incorrectly gave $H_2$ as a product from an acid–base reaction.

**(e) (i)** Many candidates gave the correct overall dipole for S=C=O but then went on to give the remaining three dipoles incorrectly.

**(ii)** Candidates frequently found it difficult to convert all of the physical quantities to consistent units. Many answers gave the identity as arsenic or some other element and clearly had not used Table 1.2 to identify the compound.